


```
// build a BST
public static Node insert(Node root, int val){
    if(root == null){
        root = new Node(val);
        return root;
    }
    if(val < root.data){
        root.left = insert(root.left, val);
    }else{
        root.right = insert(root.right, val);
    }

    return root;
}
```

```
int[] values = {8, 5, 3, 1, 4, 6, 7, 10, 11, 14};
Node root = null;

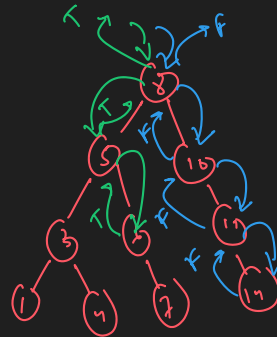
for(int i=0; i<values.length; i++){
    root = insert(root, values[i]);
}

inOrder(root);
System.out.println();
```

Q Search in a BST

$u = 1$

```
public static boolean searchInBST(Node root, int target){
    if(root == null){
        return false;
    }
    if(target < root.data){
        return searchInBST(root.left, target);
    }else if(root.data < target){
        return searchInBST(root.right, target);
    }else{
        System.out.println(root.data);
        return true;
    }
}
```



$(11) \rightarrow f$